

Project Planning and Management Report

Artificial Intelligence-Driven Diabetics Detection from Multimodal Dataset: A Pilot Study in Bangladesh

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Course: Project Planning and Management

Document Type: Comprehensive Project Charter & Management Plan

Date: September 9, 2026

Part 1: Project Definition

1.1 Project Objective

The primary objective of this project is to develop, validate, and evaluate an Artificial Intelligence (AI)-driven multimodal framework for early diabetes detection tailored to the demographic and clinical context of Bangladesh. By integrating diverse data modalities—such as clinical risk factors, demographic profiles, and non-invasive diagnostic indicators—the project aims to achieve high diagnostic accuracy in resource-constrained healthcare settings within a 12-month timeframe.

1.2 Deliverables

- **Curated Multimodal Dataset:** A anonymized dataset containing clinical, demographic, and diagnostic data collected from local pilot clinics.
- **Trained AI Model Pipeline:** A machine learning/deep learning model optimized for multimodal fusion and early diabetes risk prediction.
- **Web/Mobile Application Prototype:** A lightweight user interface for clinicians and health workers to input patient parameters and view risk assessments.
- **Pilot Evaluation & Research Report:** A detailed technical and clinical validation report summarizing model performance, deployment feasibility, and user feedback.
- **Ethical & Governance Documentation:** Institutional Review Board (IRB) approvals and data safety protocols.

1.3 Key Stakeholders

Stakeholder Group & Role / Responsibility	
Project Sponsor / University	& Provides financial backing, academic oversight, and hardware infrastructure.
Project Manager & AI Team	& Responsible for model development, data processing, and project execution.
Clinical Partners & Doctors	& Provide expert annotations, ground-truth labels, and clinical validation.
Patients / Local Community	& Primary beneficiaries and data subjects participating in the pilot study.
Regulatory Bodies (BMRC / DGHS)	& Oversee ethical compliance, health data safety, and local deployment policies.

Part 2: Project Success Criteria

2.1 Technical Success Criteria

- Achieve a predictive Accuracy $\geq 88\%$ and an Area Under the ROC Curve (AUROC) ≥ 0.90 on hold-out validation sets.
- Ensure inference latency remains under 3 seconds on standard computer hardware or mobile devices.
- Successfully implement a robust multimodal fusion architecture handling missing data fields efficiently.

2.2 Research Success Criteria

- Publication of research findings in a reputable peer-reviewed journal or high-impact IEEE/Springer conference.
- Novel contribution to multimodal healthcare datasets specifically addressing South Asian demographic nuances.

2.3 Clinical Success Criteria

- Achieve a clinical sensitivity of $\geq 90\%$ to minimize false negative rates in high-risk patients.
- Obtain a usability rating above 80% (System Usability Scale) from participating medical practitioners during pilot testing.

Part 3: Project Constraints

- **Time Constraint:** Strict 12-month project timeline driven by academic submission deadlines and funding periods.
- **Cost Constraint:** Fixed budget allocation covering hardware, participant compensation, field surveys, and software infrastructure without overrun allowances.
- **Scope Constraint:** The study is restricted to a pilot population of 1,500 subjects across selected urban and peri-urban health centers in Bangladesh.
- **Data Constraint:** Challenges associated with imbalanced class distributions, incomplete patient medical histories, and non-standardized diagnostic record formats.
- **Ethical Constraint:** Mandatory compliance with the Bangladesh Medical Research Council (BMRC) ethical guidelines, requiring informed consent in Bengali and strict patient data anonymization.

Part 4: Project Environment Analysis

4.1 Internal Factors

- **Technical Competence:** Strong expertise in machine learning, data engineering, and software development within the core team.
- **Infrastructure:** Access to university GPU clusters and computational labs.
- **Resource Allocation:** Limited dedicated human resources requiring efficient workload management.

4.2 External Factors

- **Regulatory Environment:** Bureaucratic processes involved in obtaining ethical standard approvals from medical boards.
- **Healthcare Infrastructure:** Inconsistent internet connectivity and varying technical literacy among field healthcare staff in pilot centers.
- **Socio-Cultural Factors:** Patient hesitation regarding digital health record collection, requiring clear communication in native Bengali.

Part 5: Project Management Plan

5.1 Timeline & Work Breakdown Structure (WBS)

Phase & Key Tasks & Timeline
Phase 1 & Requirements gathering, IRB ethical approval, protocol design & Months 1–2
Phase 2 & Data collection from pilot clinic sites and pre-processing & Months 3–5
Phase 3 & Model design, multimodal algorithm training, and optimization & Months 6–8
Phase 4 & System interface development, system integration, pilot deployment & Months 9–10
Phase 5 & Clinical validation, user feedback analysis, final report writing & Months 11–12

5.2 Risk Management Matrix

Risk Event & Likelihood & Impact & Mitigation Strategy
Delay in IRB Ethical Approval & High & High & Submit applications early and follow up actively with board members.
Incomplete or Noisy Data Collection & Medium & High & Implement strict data input validation scripts at collection points.
Hardware / Compute Failures & Low & Medium & Utilize cloud backup instances and localized compute checkpointing.
Low User Adoption by Clinicians & Medium & High & Conduct early user-centric UI workshops and training sessions.

5.3 Communication Plan

- **Internal Team Meetings:** Weekly virtual stand-ups to track sprint progress and technical blockers.
- **Advisory Meetings:** Bi-weekly progress updates with academic supervisors and medical advisors.
- **Stakeholder Briefings:** Monthly summary reports sent to funding bodies and health clinic administration.

5.4 Quality Management Plan

- **Data Verification:** Dual-checking patient records against gold-standard laboratory blood tests (e.g., HbA1c, OGTT).
- **Code and Architecture Audits:** Peer code reviews, version control discipline via GitHub, and rigorous k-fold cross-validation routines to prevent data leakage and overfitting.